# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** $(x+3)(x+4)$ equals: 1

**(a)** $2x+7$ **(b)** $x^2+12x+7$ **(c)** $x^2+7x+12$ **(d)** $x^2+7$

**2.** A parallelogram has base 20 cm and height 12 cm. If the base is changed to 16 cm (same area, other side), the corresponding height (in cm) is: 1

**(a)** 14 **(b)** 18 **(c)** 15 **(d)** 224

**3.** A pyramid has 9 vertices. The number of its edges is: 1

**(a)** 16 **(b)** 18 **(c)** 10 **(d)** 8

**4.** The median of 49, 15, 16, 12, 98, 66, 92, 32 is: 1

**(a)** 98 **(b)** 47.5 **(c)** 40.5 **(d)** 33

**5.** If $a=12$ and $b=3$, the value of $4a-6b+10$ is: 1

**(a)** 80 **(b)** 40 **(c)** 44 **(d)** 36

**6.** A rhombus has side 26 cm and one diagonal 20 cm. Its area is: 1

**(a)** 240 cm² **(b)** 520 cm² **(c)** 104 cm² **(d)** 480 cm²

**7.** All 12 faces of a polyhedron are triangles. The number of its edges is: 1

**(a)** 24 **(b)** 18 **(c)** 12 **(d)** 36

**8.** The mean of 5 numbers is 30. The number to be added so that the mean becomes 25 is: 1

**(a)** 0 **(b)** 25 **(c)** 5 **(d)** -5

**9.** A rectangle has length $(4x+1)$ cm and breadth $(3x+6)$ cm. Its perimeter (cm) is: 1

**(a)** $7x+7$ **(b)** $14x+7$ **(c)** $14x+14$ **(d)** $12x^2+6$

**10.** A path 3 m wide runs around the outside of a 30 m by 18 m rectangular garden. The area of the path is: 1

**(a)** 36 m² **(b)** 864 m² **(c)** 288 m² **(d)** 324 m²

**11.** A polyhedron has 32 faces and 90 edges. Its number of vertices is: 1

**(a)** 60 **(b)** 124 **(c)** 58 **(d)** 120

**12.** The mean of the first 11 multiples of 5 (that is 5, 10, ..., 55) is: 1

**(a)** 60 **(b)** 27 **(c)** 30 **(d)** 33

**13.** If $x + y = 10$ and $xy = 21$, then $x^2 + y^2$ equals: 1

**(a)** 142 **(b)** 100 **(c)** 79 **(d)** 58

**14.** A right triangle has hypotenuse 29 cm and one leg 20 cm. Its area is: 1

**(a)** 420 cm² **(b)** 210 cm² **(c)** 304 cm² **(d)** 290 cm²

**15.** A cube of side 6 cm is painted all over and cut into 1 cm cubes. The number of cubes with no face painted is: 1

**(a)** 64 **(b)** 208 **(c)** 16 **(d)** 96

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** A two-digit number has tens digit $a$ and ones digit $b$, and $a - b = 2$. Show that the number minus its reverse is 18. 2

**17.** A square garden has a diagonal of 18 m. Find its area and side length (leave the side as a surd). 2

**18.** A cuboid is 14 cm long, 8 cm wide and 5 cm high. Find the sum of lengths of all its edges and the number of edges. 2

**19.** The mean marks of 20 boys is 45 and of 30 girls is 55. Find the mean mark of the whole class of 50. 2

**20.** Expand and simplify $4(x + 3) + 2(x - 6)$. 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** The sides of a triangle are $(2x + 4)$ cm, $(5x + 2)$ cm and $(3x + 6)$ cm. Find its perimeter, and its value when $x = 3$. 3

**22.** A rectangular park is 50 m by 36 m. A path 3 m wide runs inside along its boundary. Find the area of the path. 3

**23.** All 18 faces of a polyhedron are triangles. Find the number of edges and vertices. 3

**24.** The mean of 5 numbers is 35. If one number 36 is replaced by 46, find the new mean and the change in the mean. 3

OR

If $x + y = 13$ and $xy = 40$, find $x^2 + y^2$. 3

**25.** A parallelogram has base 30 cm and height 12 cm. Another side of 18 cm has a corresponding height; find that height and the area. 3

OR

A Sierpinski triangle starts with a triangle of area 320 cm². Find the shaded area and the number of shaded triangles at stage 2. 3

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## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** A prism and a pyramid both have a hexagonal base. Find the difference between the number of 5

(i) faces

(ii) vertices

(iii) edges of the prism and the pyramid.

**OR**

The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 20, 25, 31, 35, 30, 24. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

**27.** Expand and simplify $(x+2)(x+7) - x(x+3)$. Find its value at $x = 4$. 5

**OR**

A figure is made of a rectangle 24 cm by 12 cm with a right triangle of base 24 cm and height 10 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

## END OF QUESTION PAPER — Total: 50 Marks

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